OTH REGENSBURG

Praktikum Regelungstechnik (RTP)

Versuch T5:
RetBot, distance control

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1 Preparations

1.1. (aV) P controller in the time domain

(aV) What is the equation that describes a digital P controller in the time domain? [1]

$$x_a(t) = K_P \cdot x_e(t) \tag{1.1}$$

1. $x_e(t)$: Input variable of the P controller
2. K_p: Proportional coefficient
3. $x_a(t)$: Control output of the P controller

1.2. (bV) Transfer function of open loop P controller

(bV) What is the transfer function of the open loop with P controller?
Which characteristic shows $G_o(s)$? [1]

The transfer function $G_o(s)$ is obtained using the block diagram determined in the fourth experiment as follows:

$$G_o(s) = K_P \cdot \frac{-K_I}{T_1 \cdot s^2 + s} \tag{1.2}$$

$G_o(s)$ exhibits IT1 behavior.

1.3. (cV) Transfer function of distance control with P controller

(cV) What is the transfer function of the distance control with P controller?
Which characteristic shows $G_W(s)$? [1]

By inserting 1.2, we obtain the following for $G_W(s)$:

$$G_W(s) = \frac{G_o(s)}{1 + G_o(s)} = \frac{1}{1 + \frac{-T_1}{K_P \cdot K_I} \cdot s^2 - \frac{1}{K_P \cdot K_I} \cdot s} \tag{1.3}$$

The closed-loop system $G_W(s)$ exhibits PT_2 behavior.

1.4. (dV) Reasonable damping ratio D for the closed control loop

(dV) Select a reasonable damping ratio D for the closed control loop and calculate the controller parameter K_P so that the desired/selected damping ratio is achieved. [1]

The damping ratio is set to D = 1 (aperiodic limit case) to ensure a fast, overshoot-free settling. This allows the robot to reach the target distance as quickly as possible without having to move backward. The standard mathematical form of a PT2 system is:

$$G_s = \frac{K_P}{\frac{1}{\omega_0^2} \cdot s^2 + \frac{2D}{\omega_0 \cdot s} + 1} \tag{1.4}$$

Rearranging 1.4 yields:

$$\frac{1}{\omega_0^2} = -\frac{T_1}{K_P \cdot K_I} \rightarrow \omega_0 = \sqrt{-\frac{K_P \cdot K_I}{T_1}} \tag{1.5}$$

$$\frac{2D}{\omega_0} = -\frac{1}{K_P \cdot K_I} \rightarrow \omega_0 = -K_P \cdot K_I \cdot 2 \cdot D \tag{1.6}$$

If we now set Equation 1.6 equal to 1.5, we can solve for K_P.

$$\sqrt{-\frac{K_P \cdot K_I}{T_1}} = -K_P \cdot K_I \cdot 2 \cdot D \rightarrow K_P = \frac{-1}{4 \cdot D^2 \cdot T_1 \cdot K_I} \tag{1.7}$$

Using the previously determined parameters (mean value) K_I and T_1,
Left motor:

$$K_{I,L} = 0{,}437\,\mathrm{cm\,s^{-1}\,\%^{-1}}, \qquad T_{1,L} = 57{,}1\,\mathrm{ms} = 0{,}0571\,\mathrm{s} \tag{1.8}$$

Right motor:

$$K_{I,R} = 0{,}446\,\mathrm{cm\,s^{-1}\,\%^{-1}}, \qquad T_{1,R} = 52{,}2\,\mathrm{ms} = 0{,}0522\,\mathrm{s} \tag{1.9}$$

the following values are obtained:
Left motor:

$$K_{P,L} = \frac{-1}{4 \cdot 1^2 \cdot 0.0571 \cdot 0.437} = -9{,}99\,\%\,\mathrm{cm^{-1}} \approx -10\,\%\,\mathrm{cm^{-1}} \tag{1.10}$$

Right motor:

$$K_{P,R} = \frac{-1}{4 \cdot 1^2 \cdot 0.0522 \cdot 0.446} = -10{,}77\,\%\,\mathrm{cm^{-1}} \approx -11\,\%\,\mathrm{cm^{-1}} \tag{1.11}$$

2 Experiment T5: RetBot, distance control

2.1. e) Simulink simulation of the closed control loop

(e) Simulate the idealized behavior of the closed control loop in Matlab/Simulink using the controller parameter K_P calculated in the preparation by using the setup from previous experiment. [1]

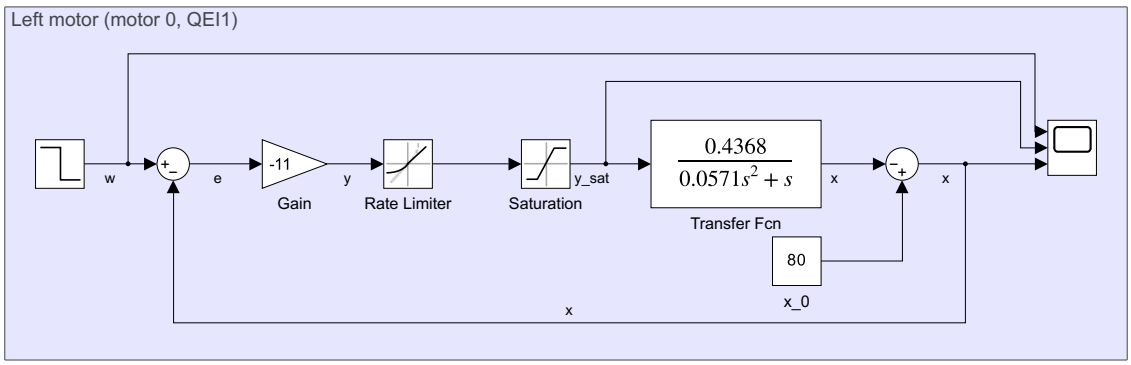


Abbildung 2.1.: Simulink model of the closed distance-control loop with the added P controller, used for the idealized simulation.

Figure 2.1 shows the Simulink model used for the simulation of the closed control loop. The model is based on the one from the previous experiment, with the addition of a P controller block. The parameters for the simulation are set according to the values calculated in section 1.4.

2.2. f) Reference, manipulated and controlled variable

(f) Plot the reference variable, the manipulated variable, and the controlled variable via Matlab (clean graphs, no screenshots from the scope). Is the damping level you selected correct? Interpret the signal curves. [1]

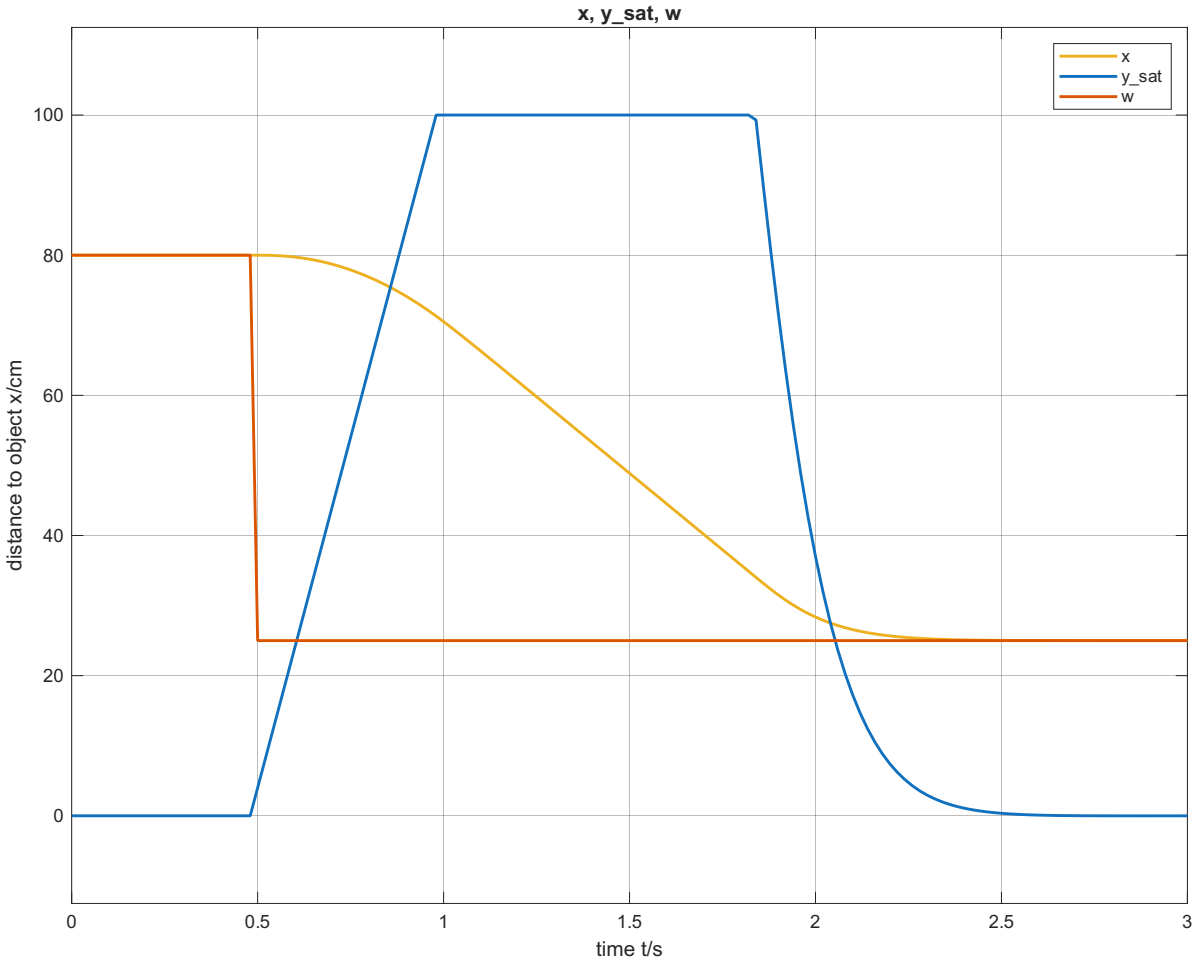


Abbildung 2.2.: Simulated step response of the closed control loop for $D = 1$: reference variable w, input variable x_a (saturated to $\pm 100\,\%$), and controlled variable x.

- **1. Evaluation of the damping:** The selected damping proves to be well-suited. The simulation shows that the controlled variable x (actual distance) rapidly converges to the reference variable w (setpoint distance) without significant overshoot or instability. The system exhibits the desired aperiodic transient response, which is ideal for distance control to avoid collisions.
- **2. Interpretation of the Signal Curves:**
  - **Reference variable $\mathbf{w(t)}$:** It represents a step change from 80 cm to 25 cm.
  - **Manipulated variable $\mathbf{x_a(t)}$:** At the start, the control error is at its maximum. The P controller theoretically calculates an extremely high manipulated variable from this. However, since the actual EV3 driver is limited to $\pm 100\,\%$, the motor moves to its initial full saturation (full throttle"). Only damped by the rate limiter, the data shows that the manipulated variable x_a does not drop significantly until shortly before reaching the target. This marks the transition from saturation to the linear control region, in order to bring the robot to a stop at the target value.

  - **Controlled variable x(t):** The distance initially decreases quadratically then it transitions to a nearly linear stretch before slowing in the final approach, since the motor runs at a constant maximum speed. As soon as the manipulated variable x_a decreases, the curve flattens out, and the robot asymptotically approaches the target value of 25 cm. Thanks to the integrating behavior of the open loop (the I-part of its IT1 characteristic), the final value is reached without steady-state error.

- **3. Necessary nonlinear element:** To correctly model the real-world conditions in the simulation, a saturation element must be added. As calculated above, the linear P controller initially outputs values well above 100 %. The saturation element limits the manipulated variable x_a in the model to the physically possible range $[-100\,\%, 100\,\%]$, which corresponds to the actual behavior of the motor driver.

2.3. g) P-controller implementation on the LEGO robot

(g) Implement the P controller for the LEGO robot - see 3.7 for the existing software-based control loop, that can be parameterized by using following commands using Serial.ICE:

- *'1': set* K_P
- *'2': set* K_I
- *'3': set* K_D
- *'4': set desired distance/cm*
- *'5': set* K_0
- *'6': set* K_1

Show the correct implementation by Serial.ICE commands and received data. [1]

For the measurement, the following values were transmitted to the LEGO robot via Serial.ICE:

Listing 2.1: Serial.ICE commands sent to the robot

```
r       -> QEI reset
m0      ->
1-11    -> set K_P to "-11"
20      -> set K_I to "0"
30      -> set K_D to "0"
425     -> set desired distance to "25" cm
50.99   -> set K_0/Motor0 to "0.99" rotations for each rotation
61      -> set K_1/Motor1 to "1" rotation for each rotation
```

An example of the received data from the robot after sending the commands is shown in Listing 2.2.

Listing 2.2: Serial.ICE commands received

```
QEI-Reset done
Motor-state: 0
PID-Kp: -11.000000
```

```
PID-Ki: 0.000000
PID-Kd: 0.000000
Desired-dist/cm: 25.000000
K0 set to 0.990000
K1 set to 1.000000
fsample=500Hz
Time/ms;QEI1/steps;QEI2/steps;d/cm;Mot0/%
   0;    0;    0; 81.0;  3
   2;    0;    0; 81.0;  3
   4;    0;    0; 81.0;  3
   6;    0;    0; 81.0;  3
   8;    0;    0; 81.0;  3
  10;    0;    0; 81.0;  3
```

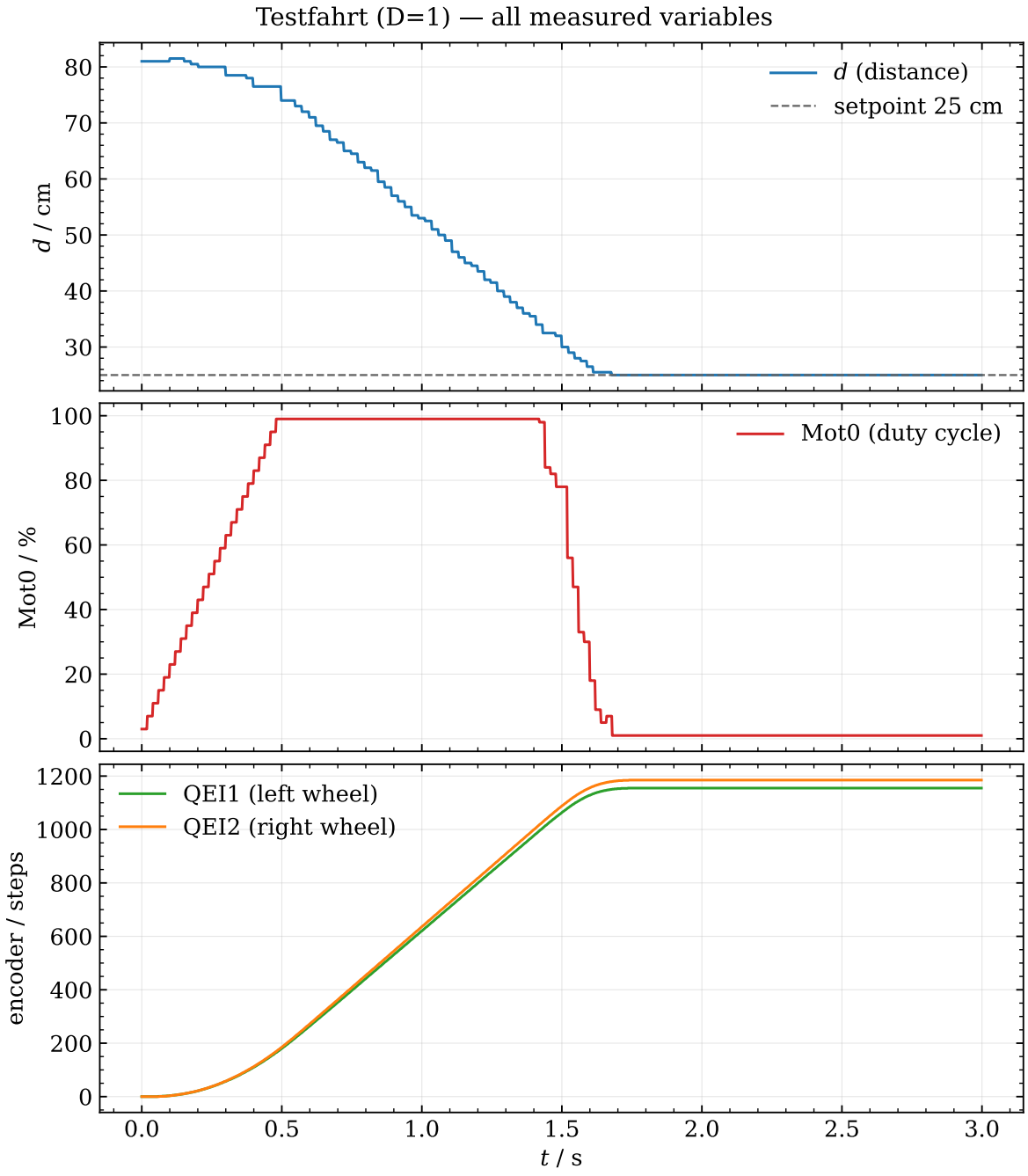
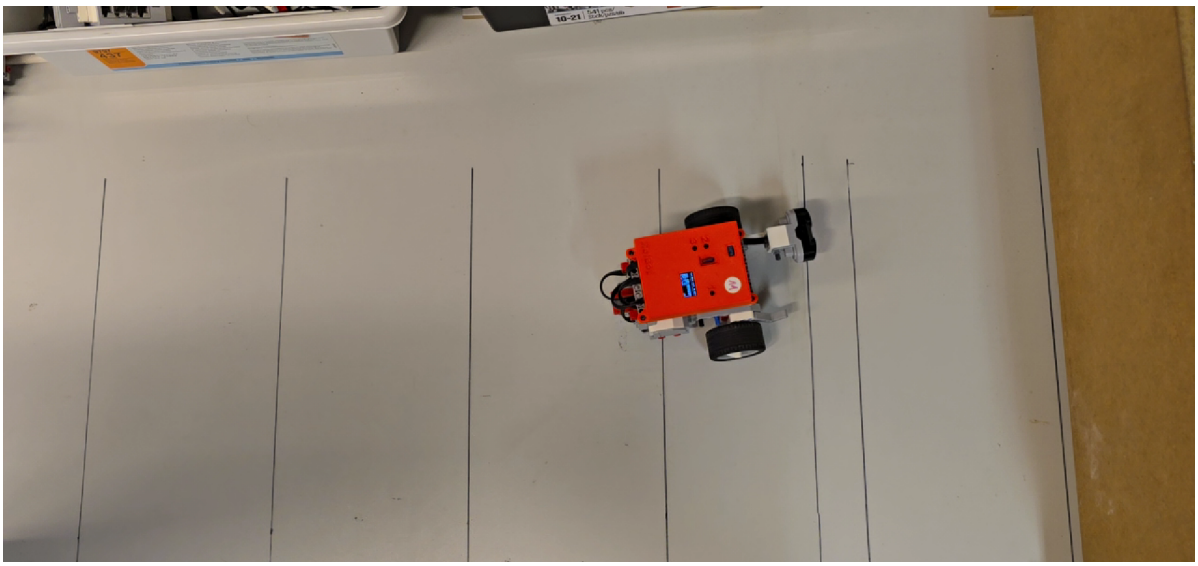


Abbildung 2.3.: Recorded test-drive ($D = 1$) logged via Serial.ICE: distance d to the obstacle (top), motor-0 duty cycle Mot0 (middle), and the two wheel-encoder counts QEI1/QEI2 (bottom) over time.

2.4. h) Video: robot approaching an obstacle

(h) Upload a video showing the robot moving towards an obstacle. [1]



https://youtu.be/gG7SXy00EJQ

Abbildung 2.4.: Video of the RetBot driving towards an obstacle (click the linked thumbnail to play).

2.5. i) Recorded test-drive vs. simulation

(i) Read the recorded data of your test-drive using Serial.ICE and compare it to the simulated results - distance, travelled distance and duty-cycle of motor 0! This has to be done in one graph after importing the data according to the previous experiment! [1]

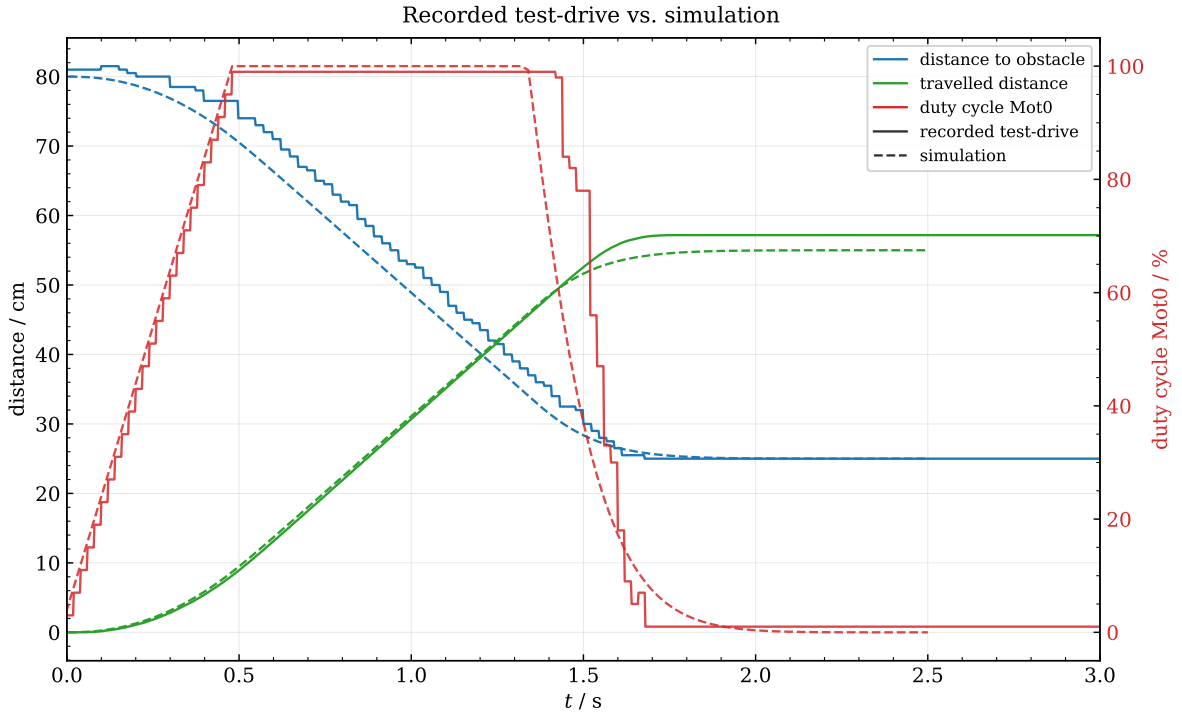


Abbildung 2.5.: Recorded test-drive (solid) versus simulation (dashed): distance to the obstacle, travelled distance, and motor-0 duty cycle. The simulation is shifted by −0,5 s onto the recording's time base.

Comparison and error analysis. Measurement and simulation are nearly identical once the simulation is shifted by −0,5 s onto the recording's time base. Both reach the setpoint with zero steady-state error (final distance 25,0 cm in either case), which confirms the integral action of the open loop. The controlled variable settles to within 1 cm of the target at 1,61 s (measurement) and 1,64 s (simulation) with a transient RMS deviation of 2,9 cm (maximum 5,8 cm just after the step). The travelled distance ends at 57,2 cm from the encoder odometry against 55,0 cm in the simulation, a deviation of 2,2 cm (4 %) that is explained by the slightly larger initial distance of the real run 81 cm vs. 80 cm and minor wheel slip. The duty cycle saturates at 100 % from 0,48 s in both cases; the real motor stays saturated about 0,1 s longer and then drops more abruptly and in discrete steps, reflecting the sampling dead-time and quantisation that the idealized model omits.

Stepped (quantised) curves. Unlike the continuous simulation, the recorded signals take on a visible staircase shape that stems from the digital nature of the measurement chain. The ultrasonic sensor resolves the distance d only in steps of 0,5 cm, so its trace advances in discrete treads instead of a smooth line. The motor duty cycle Mot0 is reported as integer percent (1 % resolution), and because the rate limiter only allows a small change per sampling cycle, it climbs as a staircase rather than a ramp. On top of that, the controller runs at a fixed sampling rate of 500 Hz, so every value is held constant between samples (zero-order hold). The wheel encoders, counting up to ∼1170 steps, are fine enough to appear practically smooth at this scale.

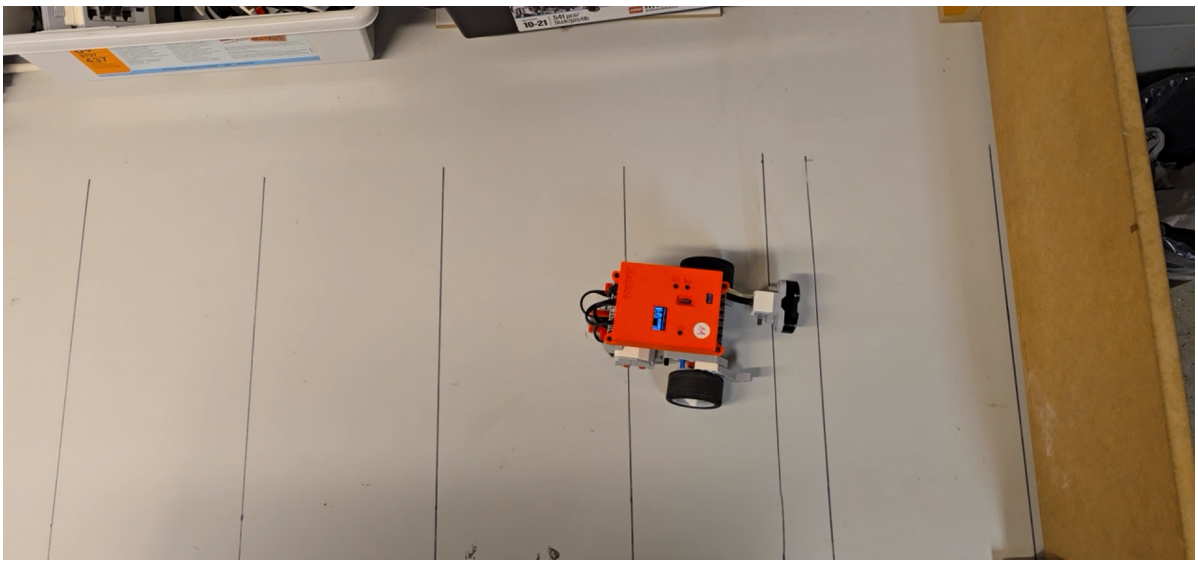
2.6. j) Critical gain and rate-limit-induced oscillation

(j) Increase K_P, check the influence of the rate-limit-factor in a way that the robot shows periodic oscillations - write down this critical value(s) and upload a video. [1]

Critical gain. To force the robot into oscillation, the damping ratio is lowered from $D = 1$ to $D = 0.5$, which quadruples the magnitude of the proportional gain. Evaluating Equation 1.4 for the right motor gives the critical value

$$K_{P,R} = \frac{-1}{4 \cdot 0.50^2 \cdot 0.0522 \cdot 0.446} = -42{,}95\,\%\,\mathrm{cm}^{-1}, \tag{2.1}$$

and $K_P = -42$ was set on the robot (Listing 2.3). At this gain the closed loop is strongly underdamped: instead of approaching the 25 cm setpoint aperiodically, the robot overshoots and swings back and forth around it.



`https://youtu.be/cy5t8bU4MxE`

Abbildung 2.6.: Video of the RetBot oscillating around the 25 cm target distance at the critical gain $K_P = -42$ (click the linked thumbnail to play).

The following Serial.ICE commands were sent for this oscillation test-drive:

Listing 2.3: Serial.ICE commands sent for the oscillation test ($K_P = -42$)

```
r
m0
1-42
20
30
425
50.99
61
```

Influence of the rate-limit factor. The rate-limit factor caps how fast the motor duty cycle may change per sampling cycle. Its main purpose is to prevent wheel slip — and the resulting odometry errors — when the robot drives at full speed, but it also acts like an additional phase lag (an effective dead-time) in the control loop. Together with the discrete sampling at 500 Hz, this lag reduces the phase margin so far that, at the high gain above, the robot no longer settles but enters a sustained, nearly periodic oscillation (a limit cycle). The oscillation is therefore not perfectly harmonic: its amplitude and period are governed by the dead-time and the rate limiter rather than by the linear PT_2 model alone.

A Appendix Frontmatter

A.1. Declarations

1. Wir erkläre hiermit, dass wir dieses Dokument selbstständig verfasst, noch nicht anderweitig für Prüfungszwecke vorgelegt, keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie wörtliche und sinngemäße Zitate als solche gekennzeichnet haben.

Regensburg, den 2026-05-01,

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Ort, Datum und Unterschrift

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B Appendix Code and Resources

B.1. Code used for the analysis of the Preperations

Editorial note — Source code omitted

The full source code for the analysis (a Jupyter notebook of this time only about 1000 lines and a simulink model) is ***not*** included in this appendix. Including it as a verbatim listing still caused major build instability again again and was therefore deliberately omitted.
The complete source can be obtained by either of the following means:

- By email request to `janis.graf@st.oth-regensburg.de`
- Via requesting access to the GitHub repository: `https://github.com/janisgra/RTP-T5-SoSe2026.git`

Literatur

[1] Torsten Reitmeier. *Internship in Control Engineering (RTP)*. Praktikumsversuche. 2025.